



Exploring Big Data on a Desktop

Getting Started with Hadoop

Hadoop is a large scale, open source storage and processing framework for data sets. In this article, the author sets up Hadoop on a single node, takes the reader through testing it, and later tests it on multiple nodes.

Fedora 20 makes it easy to install Hadoop. Version 2.2 is packaged and available in the standard repositories.

It will place the configuration files in `/etc/hadoop`, with reasonable defaults so that you can get started easily. As you may expect, managing the various Hadoop services is integrated with `systemd`.

Setting up a single node

First, start an instance, with name `h-mstr`, in OpenStack using a Fedora Cloud image (<http://fedoraproject.org/get-fedora#clouds>). You may get an IP like `192.168.32.2`. You will need to choose at least the `m1.small` flavour, i.e., 2GB RAM and 20GB disk. Add an entry in `/etc/hosts` for convenience:

```
192.168.32.2 h-mstr
```

Now, install and test the Hadoop packages on the virtual machine by following the article, <http://fedoraproject.org/wiki/Changes/Hadoop>:

```
$ ssh fedora@h-mstr
$ sudo yum install hadoop-common hadoop-common-native hadoop-
hdfs \
  hadoop-mapreduce hadoop-mapreduce-examples hadoop-yarn
```

It will download over 200MB of packages and take about 500MB of disk space.

Create an entry in the `/etc/hosts` file for `h-mstr` using the name in `/etc/hostname`, e.g.:

```
192.168.32.2 h-mstr h-mstr.novalocal
```

Now, you can test the installation. First, run a script to create the needed `hdfs` directories:

```
$ sudo hdfs-create-dirs
```

Then, start the Hadoop services using `systemctl`:

```
$ sudo systemctl start hadoop-namenode hadoop-datanode \
  hadoop-nodemanager hadoop-resourcemanager
```

You can find out the `hdfs` directories created as follows. The command may look complex, but you are running the `'hadoop fs'` command in a shell as Hadoop's internal user, `hdfs`:

```
$ sudo runuser hdfs -s /bin/bash /bin/bash -c "hadoop fs -ls
/"
Found 3 items
drwxrwxrwt - hdfs supergroup 0 2014-07-15 13:21 /tmp
drwxr-xr-x - hdfs supergroup 0 2014-07-15 14:18 /user
drwxr-xr-x - hdfs supergroup 0 2014-07-15 13:22 /var
```

Testing the single node

Create a directory with the right permissions for the user, `fedora`, to be able to run the test scripts:

```
$ sudo runuser hdfs -s /bin/bash /bin/bash -c "hadoop fs
-mkdir /user/fedora"
$ sudo runuser hdfs -s /bin/bash /bin/bash -c "hadoop fs
-chown fedora /user/fedora"
```

Disable the firewall and iptables and run a `mapreduce` example. You can monitor the progress at <http://h-mstr:8088/>. Figure 1 shows an example running on three nodes.

The first test is to calculate π using 10 maps and 1,000,000 samples. It took about 90 seconds to estimate the value of π to be 3.1415844.

```
$ hadoop jar /usr/share/java/hadoop/hadoop-mapreduce-
examples.jar pi 10 1000000
```

In the next test, you create 10 million records of 100 bytes each, that is, 1GB of data (~1 min). Then, sort it (~8 min) and, finally, verify it (~1 min). You may want to clean up the directories created in the process: